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Lab Task 7

(1)

FCFS (first come first serve):

P1	P2	P3	P4

0 6 14 21 24

Process	AT	CT	BT
P1	0	6	6
P2	2	14	8
P3	4	21	7
P4	6	24	3

(1) P_1:

Copleletion time = 6

$\boxed{\begin{aligned} TAT &= \text{Completion time} - \text{Arival time} \\ WT &= TAT - BT \end{aligned}}$

$TAT = 6 - 0 = 6$

$WT = 6 - 6 = 0$

(2) P_2:

$CT = 14$

$TAT = 14 - 2 = 12$

$WT = 12 - 8 = 4$

CamScanner

P_3:

CT = 21

TAT = 21 - 4 = 17

WT = 17 - 7 = 10

P_4:

CT = 24

TAT = 24 - 6 = 18

WT = 18 - 3 = 15

$$\boxed{\text{Average WT} = \frac{0 + 4 + 10 + 5}{4} = 7.25}$$

$$\boxed{\text{Average TAT} = \frac{6 + 12 + 17 + 18}{4} = 13.25}$$

S T F (Shortest Job First): (Non-preemptive)

② SJf (shortest Job first):

	P1	P2	P4	P3
0	1	6	8	12	21

Completion time = start + Burst time

TAT = Completion time - Arival time

WT = TAT - BT

Process	AT	BT	CT	TAT	WT
P1	1	5	6	5	0
P2	3	2	8	5	3
P3	5	9	21	16	7
P4	6	4	12	6	2

$$\boxed{\text{Average TAT} = \frac{5+5+16+6}{4} = 8}$$

$$\boxed{\text{Average WT} = \frac{0+3+7+2}{4} \quad 3}$$

③ SRTF (Preemptive).

P1	P2	P4	P3	P1

0 1 5 7 14 22

$$\boxed{\begin{aligned} TAT &= CT - AT \\ WT &= TAT - BT \end{aligned}}$$

Process	ArT	BT	CT	TAT	WT
P_1	0	9	22	22	13
P_2	1	4	5	4	0
P_3	2	7	14	12	5
P_4	3	2	7	4	2

$$\boxed{\text{Average TAT} = \frac{22+4+12+4}{4} = 10.5}$$

$$\boxed{\text{Average WT} = \frac{13+0+5+2}{4} = 5}$$

4) Round Robin (RR) Sheduling

Time Quantum = 3 ms

P_2	P_3	P_4	P_1

P_1	P_2	P_3	P_4	P_1	P_2	P_3	P_1	P_3	P_1

0 3 6 9 12 15 17 20 23 25 26

Process	AT	BT	CT	TAT	WT
P_1	0	10	26	26	16
P_2	1	5	17	16	11
P_3	2	8	25	23	15
P_4	3	3	12	9	6

$$\text{Average TAT} = \frac{26+16+23+9}{4} = 18.5$$

$$\text{Average WT} = \frac{16+11+15+6}{4} = 12$$

5 Round Robin (RR):

Time Quantum = 2 ms

P1	P2	P1	P3	P2	P1	P4	P3	P2

0 2 4 6 8 10 12 14 16 18

P1	P4	P3	P1

18 20 22 24 26

TAT = CT − AT

WT = TAT − BT

Process	AT	BT	TAT	CT	WT
P_1	0	12	30	30	18
P_2	2	6	16	18	10
P_3	4	8	24	28	16
P_4	6	4	16	22	12

$$\boxed{\text{Average TAT} = \frac{30+16+24+6}{4} = 21.5}$$

$$\boxed{\text{Average WT} = \frac{18+10+16+12}{4} = 14.}$$

Task 8:

Priority sheduling (Non-preemptive)

As P1 arrive at t=0

So,

P1	P2	P3	P4

0 7 11 17 22

Process	AT	BT	CT	TAT	WT	Priority
P_1	0	7	7	7	0	3
P_2	1	4	11	10	6	1
P_3	2	6	17	15	9	2
P_4	3	5	22	19	14	4

Average Turnaround Time $= \frac{7+10+15+19}{4} = 12.75$

Average waiting time $= \frac{0+6+9+14}{4} = 7.25$

Task 7:- Priority sheduling (Preemptive)

P1	P2	P1	P3
0	2	4	11	16

Process	AT	BT	Priority	CT	TAT	WT
P_1	0	8	2	11	11	3
P_2	1	3	1	4	3	0
P_3	2	5	3	16	14	9

(1)

$$\text{Average TAT} = \frac{11 + 3 + 14}{3} = 9.33$$

$$\text{Average WT} = \frac{3 + 0 + 9}{3} = 4.00$$

(2